Electromagnetic Induction (Faraday's Law)

1. Definition

Electromagnetic induction is the phenomenon of producing an electromotive force (EMF) in a conductor when the magnetic flux linked with it changes. If the circuit is closed, an induced current flows through it.

2. Faraday's Laws of Electromagnetic Induction

First Law: Whenever the magnetic flux linked with a circuit changes, an EMF is induced in the circuit.

Second Law: The magnitude of the induced EMF is directly proportional to the rate of change of magnetic flux linked with the circuit.

Formula:

$\varepsilon = -N(\Delta\Phi/\Delta t)$

Where:

- ε = induced EMF (volts)
- N = number of turns in the coil
- $\Delta\Phi$ = change in magnetic flux (weber)
- Δt = time interval (seconds)

The negative sign represents Lenz's law.

3. Magnetic Flux

Magnetic flux is the measure of the magnetic field passing through a given surface.

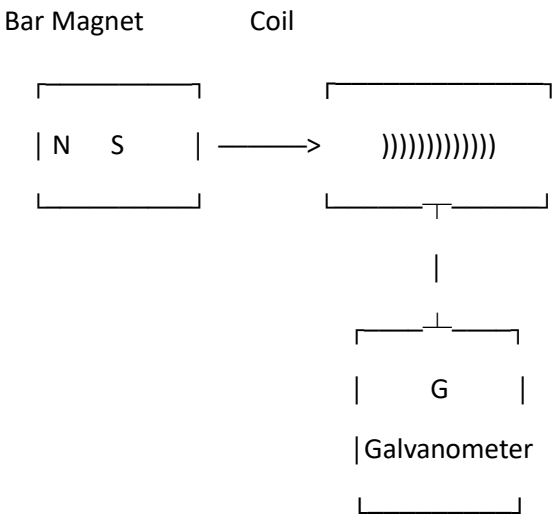
$\Phi = BA \cos\theta$

Where:

- B = magnetic field strength (tesla)
- A = area of the surface (m^2)
- θ = angle between the magnetic field and the normal to the surface

4. Experimental Setup

A bar magnet is moved towards or away from a coil connected to a galvanometer. The galvanometer detects the induced current.



5. Procedure

1. Connect a coil to a galvanometer.
2. Move a bar magnet towards the coil and observe the galvanometer.
3. Keep the magnet stationary and observe the reading.
4. Move the magnet away from the coil and observe the reading.
5. Repeat the experiment by moving the magnet at different speeds.

6. Observation Table

S. No.	Action	Galvanometer observation
1	Magnet moved towards the coil	Deflection in one direction
2	Magnet kept stationary	No deflection
3	Magnet moved away from the coil	Deflection in the opposite direction
4	Magnet moved faster towards the coil	Larger deflection

7. Lenz's Law

Lenz's law states that the induced current flows in a direction such that the magnetic field produced by it opposes the change in magnetic flux that causes it.

This explains the negative sign in Faraday's law.

8. Factors Affecting Induced EMF

1. Strength of the magnetic field.
2. Number of turns in the coil.

3. Speed of relative motion between the magnet and coil.
4. Area and orientation of the coil.

9. Applications

- Electric generators
- Transformers
- Induction cooktops
- Wireless charging
- Induction motors
- Dynamic microphones

10. Result

Electromagnetic induction occurs when the magnetic flux linked with a circuit changes. The induced EMF is proportional to the rate of change of magnetic flux, as stated by Faraday's law.

$\varepsilon = -N(\Delta\Phi/\Delta t)$